

# Rational Numbers on the Number Line

Lesson 9-4

**Name:** Type your name      **Date:** Today's date      **Class:** Class period

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## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can place rational numbers, including fractions and decimals, on a number line.

**Language Objective:** I can explain my reasoning using the words rational number, fraction, decimal, and number line.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rational number

Fraction

Decimal

Number line

Equivalent

Integer



Tap any word to see what it means and a picture.

- 1 Any number that can be written as a fraction, including integers and decimals, is a

.

- 2 A number that shows part of a whole, like  $\frac{3}{5}$ , is a  .

- 3 A number that uses a point to show parts of a whole is a  .

- 4 A line with numbers in order used to place and compare values is a

.

- 5 Numbers that name the same value, like  $\frac{1}{2}$  and 0.5, are

.

6 A whole number or its opposite is an  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Where would you place  $-1\frac{1}{2}$  on the number line, and how do you decide between which two integers it falls?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Rational number** — Any number you can write as a fraction. This includes fractions, decimals, and integers.

*Número racional — Cualquier número que puedes escribir como fracción. Incluye fracciones, decimales y enteros.*

- ☐ **Fraction** — A number that shows part of a whole, like  $\frac{3}{4}$ .

*Fracción — Un número que muestra una parte de un todo, como  $\frac{3}{4}$ .*

- ☐ **Number line** — A straight line with numbers spaced out evenly.

*Recta numérica — Una línea recta con números separados de forma pareja.*

- ☐ **Integer** — Whole numbers and their opposites, like -2, -1, 0, 1, 2.

*Número entero — Números enteros y sus opuestos, como -2, -1, 0, 1, 2.*

- ☐ **Decimal** — A number with a dot, like 0.5, that shows a part less than one.

*Decimal — Un número con un punto, como 0.5, que muestra una parte menor que uno.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**-1  $\frac{1}{2}$  is between \_\_\_\_ and \_\_\_\_.**

*-1  $\frac{1}{2}$  esta entre \_\_\_\_ y \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** All of these are rational numbers because each one can be \_\_\_\_.

**Evidence:** Student explains a rational number is any number that can be written as a fraction, including integers and decimals.

**Reasoning:** This evidence connects the definition of rational number to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **-1 1/2 is between \_\_\_\_ and \_\_\_\_.**

*-1 1/2 esta entre \_\_\_\_ y \_\_\_\_.*

- **It is closer to \_\_\_\_ because \_\_\_\_.**

*Esta mas cerca de \_\_\_\_ porque \_\_\_\_.*

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*